Physics 5: Speed

CHEAT SHEET

@nova.xplains

What is speed?

$$v = \frac{s}{t}$$

v velocity = speed
s distance
t time

v = s ÷ t: speed = distance divided by time

v = velocity, s = distance, t = time

Example: 10 m in 10 s = 1 m/s

Two units



km/h: kilometers in one hour (road signs, car)

m/s: meters in one second (sprint, falling, braking)

km/h › m/s: ÷ 3.6 · m/s › km/h: × 3.6

Why 3.6?



1 km = 1000 m and 1 h = 3600 s

1 km/h = 1000 m ÷ 3600 s ≈ 0.28 m/s

30 m/s × 3.6 = 108 km/h, faster than 100 km/h!

Snail to light

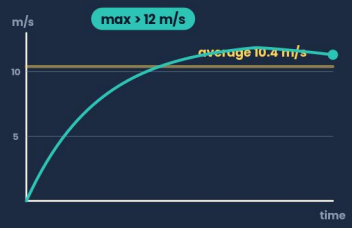


Snail ≈ 1 mm/s · walking ≈ 5 km/h ≈ 1.4 m/s

Usain Bolt: 100 m in 9.58 s, top ≈ 45 km/h · cheetah ≈ 100 km/h

Plane ≈ 900 km/h · sound ≈ 340 m/s · light ≈ 300,000 km/s

Average vs. instantaneous



Average speed = whole distance ÷ whole time

Bolt: 100 m ÷ 9.58 s ≈ 10.4 m/s on average

Instantaneous speed = speed at one moment (speedometer)

Time and distance



s = v · t

t = s ÷ v

v = s ÷ t

s = v · t and t = s ÷ v

3 km by bike at 15 km/h: t = 3 ÷ 15 = 0.2 h = 12 min

50 km/h ≈ 14 m/s: 1 s on the phone = 14 m blind!

BONUS Quiz: 72 km/h ÷ 3.6 = 20 m/s (answer B). Next episode: acceleration.

Sources: World Athletics (100 m world record 9.58 s, Berlin 2009); SI units (BIPM); physics textbooks

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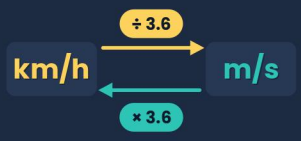
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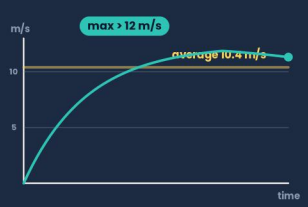


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